

CTQ - Clinical Trial Qualifier

AI-Powered Clinical Trial Matching and Eligibility Analysis for Indian Patients

This report explains the complete, final system: what it does, how every technical part works (with a worked example at each step), a full guided tour of the website with live screenshots of the current build, and the measured results of a live evaluation run.

One-line summary: you enter a patient (structured form, or unstructured notes / PDF / photo / CSV / Excel upload), a DistilBERT ML NER turns free text into structured facts, CTQ screens that patient against every trial in its CTRI-format database with semantic search + a deterministic rule engine, and returns one of three explained verdicts per trial: Potentially Eligible, Not Eligible, or Insufficient Information. A Model Evaluation page scores the whole pipeline against ground-truth labels.

Live result quoted throughout this report: 10 synthetic patients x 21 labelled patient-trial pairs → Accuracy 100%, Precision 100%, Recall 100%, F1 100% (TP 11, FP 0, FN 0, TN 10, zero pairs excluded) - reproduced from the running app on 28 Sep 2026. Scoring against only each patient's top-5 best-matching trials also gives 100% (13 pairs scored, 8 reported as outside the ranked scope).

1. What the project does

Clinical trials publish eligibility criteria as long free-text lists: age windows, lab cut-offs ("HbA1c between 7.5 and 10.5 percent"), medication requirements ("on stable metformin for at least 3 months"), exclusion clauses. Patients and clinicians rarely have the time to read every trial's criteria. CTQ automates exactly that reading.

The system performs five jobs:

1. Understand the patient. Three input paths on one page: the **Structured Data** tab (60+ guided form fields, or a structured CSV upload, with a cohort mode for many patients), the **Unstructured Data** tab (clinical notes as PDF, photo, CSV, Excel, TXT or JSON), and the **Hybrid** tab (a structured CSV plus the notes, merged into one authoritative profile per patient - CSV fields win, ML-extracted facts fill every gap). Non-English notes (Hindi and 12 other scripts) are auto-translated to English before extraction. Free text is converted to structured facts by the DistilBERT ML NER - no hand-written regex discovery.

2. Find candidate trials. A structured pre-filter annotates (and only drops clearly impossible) trials; TOP_K_TRIALS=0 means every trial in the 32-trial database is actually checked for each patient.

3. Rank by meaning. The patient profile and each trial's criteria are embedded into vectors (all-MiniLM-L6-v2, local) and compared with cosine similarity. This is the Match Score % - 'how similar is this patient to this trial'.

4. Verify the details. A deterministic rule engine checks everything objectively checkable: numeric lab cut-offs parsed from the criterion text, age and gender windows, medication requirements and durations, comorbidity exclusions, pregnancy and lifestyle criteria. Missing inclusion-side facts produce 'Insufficient Information' rather than a guess; missing exclusion-side facts keep 'Potentially Eligible' with the gap listed.

5. Explain, export, and evaluate. Every verdict ships with itemised reasons quoting the trial's own criteria; results download as CSV, the cohort exports as Excel. The Model Evaluation page runs the same pipeline over labelled pairs and reports accuracy, precision, recall, F1 and a confusion matrix.

2. Why this design is good

Honest about uncertainty. Most systems force a yes/no answer. CTQ has a third verdict, 'Insufficient Information', and names exactly which fields are missing. In screening, a wrong confident answer is far more harmful than an honest 'need more data'.

Auditable, not a black box. Every verdict lists the exact criteria checked (e.g. 'HbA1c 8.5 satisfies inclusion criterion "HbA1c between 7.5 and 10.5"'). Because structured facts are checked with code, the reasons are reproducible - the ML only extracts, it never invents facts and can never overturn a rule-engine hard failure.

ML where ML wins, rules where rules win. Language understanding (reading a note, coping with phrasing variety) is ML's strength; safety-critical verification (a numeric cut-off, a hard exclusion) is code's strength. CTQ assigns each to the right component.

Built for the Indian context. Trials are modelled on the Clinical Trials Registry - India (CTRI), with city/state locations shown on every card, and the UI is designed to be simple enough for non-technical users.

Measurable. The Model Evaluation page has a one-click bundled sample (10 patients + 21 labels): anyone can reproduce the metrics in this report in about two minutes, no file preparation needed.

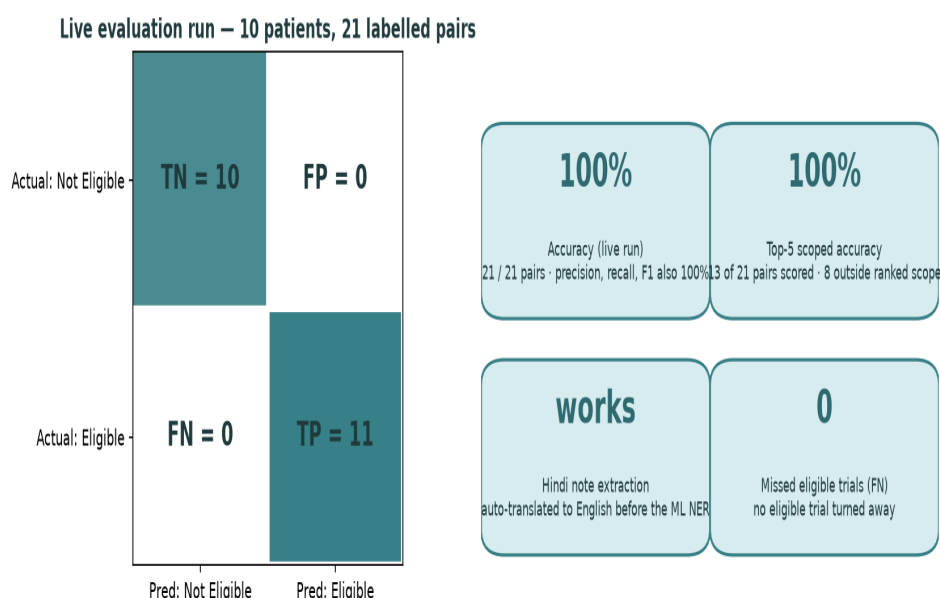
3. How accurate is it - live measured results

3.1 Live evaluation run (this exact build, 27 Sep 2026)

The bundled sample set ships with the app: 10 synthetic patients as free-text clinical notes plus 21 ground-truth labelled patient-trial pairs (11 eligible, 10 not eligible) covering diabetes, hypertension, asthma, COPD, healthy-volunteer, nephropathy, obesity and prediabetes trials. The full pipeline - ML NER, retrieval, rule engine - ran over every pair:

Metric	Value
Patients evaluated	10
Labelled pairs scored	21 (0 skipped, 0 excluded)
Accuracy	100%
Precision	100%
Recall	100%
F1 score	100%
Confusion matrix	TP 11 · FP 0 · FN 0 · TN 10

CTQ Evaluation Results



The confusion matrix above is generated from the live run's JSON output, not drawn by hand. FN = 0 means no eligible trial was ever turned away - the clinically most important property for a screening tool.

3.2 Ranking scope - accuracy against the top-K matches (new)

In real use, a coordinator reads the ranked list from the top - not all 32 trials. The Model Evaluation page therefore offers a ranking scope: All trials, Top 3, Top 5 or Top 10. Under a top-K scope, a labelled pair is scored only when that trial is among the patient's K best-matching trials (by embedding similarity); pairs outside the scope are skipped and reported next to the confusion matrix, so the denominator is always visible. On the bundled sample:

Scope	Pairs scored	Skipped (outside scope)	Accuracy	Confusion
All trials	21	0	100%	TP 11 · FP 0 · FN 0 · TN 10
Top-5 trials per patient	13	8	100%	TP 9 · FP 0 · FN 0 · TN 4
Top-3 trials per patient	11	10	100%	TP 9 · FP 0 · FN 0 · TN 2

The top-5 run keeps FN = 0 and precision = 100% while evaluating the system exactly the way its users read the results - the ranked shortlist first.

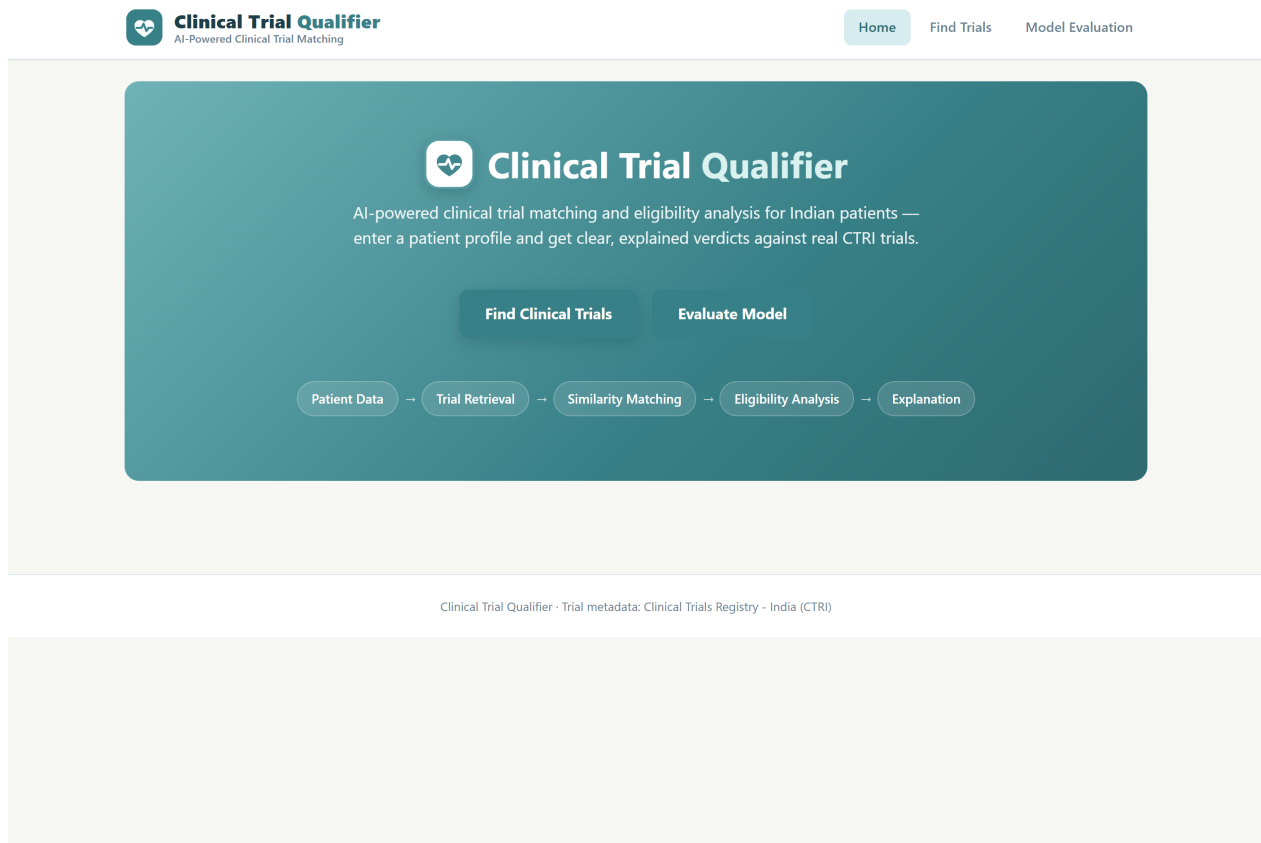
3.3 Honesty mechanism (why nothing was 'guessed')

Rows whose prediction is 'Insufficient Information' are excluded from the binary matrix and reported separately (in this run: zero). If a note genuinely lacks a fact - e.g. an asthma note without FEV1 - the matching pair degrades to 'Insufficient Information' instead of a lucky guess, which is the behaviour a screening tool should have.

Both evaluation files also exist as standalone CSVs (sample_data/evaluation_5_patients_unlabelled.csv + evaluation_5_patients_labelled.csv for a quick 5-patient run; evaluation_patients_unstructured.csv + evaluation_labels_ground_truth.csv for the 10-patient run) and can be uploaded manually on the Model Evaluation page.

4. Guided tour - how each part works (live screenshots)

4.1 Home - the front door



The hero states what the product does with two calls to action: Find Clinical Trials (the matcher) and Evaluate Model (accuracy measurement). The five pills show the pipeline a search will follow: Patient Data, Trial Retrieval, Similarity Matching, Eligibility Analysis, Explanation.

4.2 Find Trials - two input modes, one page

Patient → Trial Matcher

Enter patient information to find potentially relevant Indian clinical trials. All fields except age are optional - missing information leads to honest "Insufficient Information" verdicts rather than guesses.

Try it instantly with sample data

[Sample: Diabetes \(47F\)](#)
[Sample: Hypertension \(58M\)](#)
[Sample: Asthma \(24F\)](#)
[Sample: Healthy volunteer \(28M\)](#)

Single patient mode

Filling several patients? Add them below - each one gets its own trial check.

[+ Add another patient](#)
[↓ Download all patients \(Excel\)](#)
[↑ Upload structured CSV](#)
[Structured Data \(form\)](#)
[Unstructured Data \(photos, PDF, CSV\)](#)
[Hybrid \(structured + unstructured\)](#)

Basic Information

Patient ID

Any code, e.g. P001 - no personal names needed

Age *

Gender

State

City

Height (cm)

Weight (kg)

Disease Information

Primary disease / condition

e.g. Type 2 Diabetes

Disease duration (months)

Disease severity

Previous diagnosis

Symptoms

☐ Increased thirst

☐ Frequent urination

☐ Fatigue

☐ Blurred vision

☐ Numbness in feet

The Structured Data tab is open by default: one-click sample profiles (Diabetes 47F, Hypertension 58M, Asthma 24F, Healthy volunteer 28M), the cohort bar ('Single patient mode' / '+ Add another patient' / 'Download all patients (Excel)'), and the Unstructured Data tab switch at the top.

4.3 Structured form - medications and labs

Patient → Trial Matcher

Enter patient information to find potentially relevant Indian clinical trials. All fields except age are optional - missing information leads to honest "Insufficient Information" verdicts rather than guesses.

Try it instantly with sample data

[Sample: Diabetes \(47F\)](#)
[Sample: Hypertension \(58M\)](#)
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[Sample: Healthy volunteer \(28M\)](#)

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[Hybrid \(structured + unstructured\)](#)

Basic Information

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Any code, e.g. P001 - no personal names needed

Age *

Gender

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City

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Disease Information

Primary disease / condition

e.g. Type 2 Diabetes

Disease duration (months)

Disease severity

Previous diagnosis

Symptoms

☐ Increased thirst

☐ Frequent urination

☐ Fatigue

☐ Blurred vision

☐ Numbness in feet

The diabetes sample is loaded. Each medication row captures name, dose ('500 mg'), frequency ('twice daily') and duration in months - the details criteria like 'stable metformin for at least 3 months' need. The Laboratory Values grid holds HbA1c, glucose, BP, creatinine, eGFR, haemoglobin, WBC, platelets, ALT, AST, bilirubin, cholesterol and FEV1 % predicted. Left-blank fields become Unknown - which is exactly what produces honest 'Insufficient Information' verdicts instead of guesses.

4.4 Cohort mode - many patients, one run

**Clinical Trial Qualifier**
AI-Powered Clinical Trial Matching

HomeFind TrialsModel Evaluation

Patient → Trial Matcher

Enter patient information to find potentially relevant Indian clinical trials. All fields except age are optional - missing information leads to honest "Insufficient Information" verdicts rather than guesses.

Try it instantly with sample data

Sample: Diabetes (47F)

Sample: Hypertension (58M)

Sample: Asthma (24F)

Sample: Healthy volunteer (28M)

2 patients in the cohort

Patient 1 is the form above; the others are listed below it.

+ Add another patient

Download all patients (Excel)

Clear cohort (2 patients)

Structured Data (form)

Unstructured Data (photos, PDF, CSV)

Patient 2 of 2

X Remove

Patient ID

Age *

Gender

Primary condition

P002

Select...

e.g. Type 2 Diabetes

Disease duration (months)

Height (cm)

Weight (kg)

HbA1c (%)

eGFR

FEV1 (% predicted)

Pregnancy status

Smoking status

Select...

Select...

Alcohol use

Select...

Medical history

☐ Diabetes

☐ Hypertension

☐ Cardiovascular disease

☐ Kidney disease

☐ Liver disease

After '+ Add another patient', each added patient gets its own full form card ('Patient 2 of 2'), the counter reads '2 patients in the cohort', and the submit button becomes 'Find Trials for all 2 patients'. Results come back per patient with a ← Previous / Next patient → stepper, and the whole cohort's entered details download as one Excel file.

4.5 Results - 'Your Trial Matches'

Your Trial Matches

New search

32 trials · 10 potentially eligible · 20 not eligible · 2 need more info

Eligibility

Match score

Disease

State

Recruitment status

All

Highest match score first

All

All

All

Phase

All

CTRI/2024/01/062001

A Randomised Open Label Trial of Glimepiride Add-on to Metformin in Type 2 Diabetes Mellitus

✓ **Potentially Eligible** **Match Score: 78%** Type 2 Diabetes Mellitus Recruiting Phase III Siliguri, West Bengal

Why?

- ✓ Patient age 47 is within the required range 30-65 years
- ✓ Patient condition "Type 2 Diabetes" matches trial condition "Type 2 Diabetes Mellitus"
- ✓ HbA1c 8.2 satisfies inclusion criterion "HbA1c between 7.5 and 10.5 percent at screening"
- ✓ BMI 29.7 satisfies inclusion criterion "Body mass index between 23 and 35"
- ✓ Patient is on metformin, satisfying "On stable metformin therapy for"
- ✓ No conflict with the trial's pregnancy/breastfeeding exclusion
- ✓ Medical history shows no liver disease, matching exclusion about liver disease
- ✓ Age within 30-65 years
- ✓ Diagnosed with Type 2 Diabetes Mellitus
- ✓ HbA1c between 7.5% and 10.5%
- ✓ On stable metformin therapy for >3 months

Missing information

- ✗ eGFR value to assess renal function
- ✗ ALT level to assess hepatic function

After pressing Find Matching Trials, the backend runs the pipeline and the Results page lists every trial with its verdict badge (green Potentially Eligible, red Not Eligible, amber Insufficient Information), Match Score %, registry ID, locations and phase, plus summary counts. A 'Download results (CSV)' button sits below the list.

4.6 Every verdict is explained

All

Highest match score first

All

All

All

Phase

All

CTRI/2024/01/062001

A Randomised Open Label Trial of Glimepiride Add-on to Metformin in Type 2 Diabetes Mellitus

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 - ✓ No conflict with the trial's pregnancy/breastfeeding exclusion
 - ✓ Medical history shows no liver disease, matching exclusion about liver disease
 - ✓ Age within 30-65 years
 - ✓ Diagnosed with Type 2 Diabetes Mellitus
 - ✓ HbA1c between 7.5% and 10.5%
 - ✓ On stable metformin therapy for >3 months
 - ✓ BMI between 23 and 35
 - ✓ Not pregnant
- No known exclusion criterion detected.

Missing information

- ✗ eGFR value to assess renal function
- ✗ ALT level to assess hepatic function

View Full Trial

Each card carries green reasons-for (e.g. 'HbA1c 8.5 satisfies inclusion criterion "HbA1c between 7.5 and 10.5 percent"'), red reasons-against (e.g. 'BMI 28.4 does not satisfy "Body mass index between 30 and 45"'), and amber missing-information naming the exact fields never provided. This is the anti-black-box guarantee: the same list a clinician would produce by hand, generated in seconds.

4.7 Unstructured Data tab - drop any document

**Clinical Trial Qualifier**
AI-Powered Clinical Trial Matching

HomeFind TrialsModel Evaluation

Patient → Trial Matcher

Enter patient information to find potentially relevant Indian clinical trials. All fields except age are optional - missing information leads to honest "Insufficient Information" verdicts rather than guesses.

Try it instantly with sample data

Sample: Diabetes (47F)

Sample: Hypertension (58M)

Sample: Asthma (24F)

Sample: Healthy volunteer (28M)

Structured Data (form)

Unstructured Data (photos, PDF, CSV)

Unstructured patient data

Upload a photo of a prescription or discharge summary (JPG/PNG), a PDF, or a CSV/Excel/TXT file of medical history. The DistilBERT ML NER reads the text, converts it into structured profile fields, and every trial in the database is checked.

Choose file

No file chosen

Clinical Trial Qualifier · Trial metadata: Clinical Trials Registry - India (CTRI)

Accepts PDF, photos (OCR via RapidOCR), CSV, Excel, TSV, TXT and JSON. A text column is found by alias (notes / clinical_notes / summary / medical_history...) and a patient id column by alias (patient_id / note_id / id). The same box serves the Model Evaluation page.

4.8 Hindi clinical note - auto-translated, then extracted

Patient → Trial Matcher

Enter patient information to find potentially relevant Indian clinical trials. All fields except age are optional - missing information leads to honest "Insufficient Information" verdicts rather than guesses.

Try it instantly with sample data

Sample: Diabetes (47F)

Sample: Hypertension (58M)

Sample: Asthma (24F)

Sample: Healthy volunteer (28M)

Structured Data (form)

Unstructured Data (photos, PDF, CSV)

Hybrid (structured + unstructured)

Unstructured patient data

Upload a photo of a prescription or discharge summary (JPG/PNG), a PDF, or a CSV/Excel/TXT file of medical history. The DistilBERT ML NER reads the text, converts it into structured profile fields, and every trial in the database is checked.

Choose file hindi_note_short.csv

Extracted structured data from 1 note in "hindi_note_short.csv" (table).

1 of 1 note(s) were in a non-English script and were auto-translated to English before ML extraction (Devanagari).

H001

Patient 1 of 1

Age 48

Diabetes

576 mo

Smoke: Never

Diabetes

hba1c 8.5

egfr 80

systolic_bp 140

diastolic_bp 85

Find Matching Trials for this profile

Download extracted results (CSV)

The uploaded file contains a fully Hindi discharge note ('■■■■■ ■■■■■, ■■■■ 48 ■■■■... ■■■■■■■■■■■■ 500 ■■■■■■■■■■■■...'). CTQ detects the Devanagari script, translates the note to English via the Groq LLM, and the DistilBERT NER then extracts the complete profile - age 48, male, diabetes, Metformin, HbA1c 8.5, eGFR 80, BP 140/85. The banner reports the translation; without it, only Latin-anchored lab values ('HbA1c 8.5') would survive. Translation is behind the same rate-limit breaker as the LLM review: if Groq is unavailable the original text is used and the UI says so.

4.9 Model Evaluation - zero-preparation demo + ranking scope

Model Evaluation

Give **unstructured patient data** (notes / summaries / CSV of medical history) plus **labelled structured data** (ground truth). The ML NER converts each unstructured patient into a structured profile, every trial in the database is checked, and the predictions are scored against your labels: accuracy, precision, recall, F1 and the confusion matrix.

Sample data

No files handy? Load the synthetic sample set shipped with the app: 10 patient notes plus 21 labelled patient/trial pairs. Then just press **Run Evaluation**.

[Load bundled sample data \(10 patients, 21 labels\)](#)

1 · Unstructured patient data * (PDF / photo / CSV / Excel / TXT / JSON)

A file with clinical notes per patient — e.g. columns like `patient_id`, `notes` (CSV), one note per line (TXT), or a document of medical history. Accepted formats:

`.pdf .png .jpg .jpeg .csv .xlsx .tsv .txt .json`

[Choose file](#) No file chosen

Loaded **10** patient notes from **evaluation_patients_unstructured.csv** (sample).

2 · Labelled structured data * (ground truth)

CSV with columns `patient_id`, `trial_id`, `actual_label` — `actual_label`: 1 = Potentially Eligible, 0 = Not Eligible, 2 = Insufficient Information.

[Choose file](#) No file chosen

Loaded **21** labels from **evaluation_labels_ground_truth.csv**.

3 · Ranking scope

Which trials should accuracy be scored against? **All trials** checks every trial in the database. **Top-K** scores only each patient's K best-matching trials (by match score) — the way a coordinator actually reads the ranked list. Labelled pairs that fall outside a patient's top-K are skipped and reported.

[All trials](#)

[Top 3 trials](#)

[Top 5 trials](#)

[Top 10 trials](#)

Four numbered steps: unstructured patient file, labelled ground-truth file, ranking scope, run. The 'Load bundled sample data (10 patients, 21 labels)' button fills both inputs from the app itself - no file handling needed for a demo. The ranking-scope buttons choose between scoring all trials and scoring only each patient's top-3 / top-5 / top-10 best matches (section 3.2).

4.10 The live metrics dashboard (top-5 scope)

Model Evaluation

Give **unstructured patient data** (notes / summaries / CSV of medical history) plus **labelled structured data** (ground truth). The ML NER converts each unstructured patient into a structured profile, every trial in the database is checked, and the predictions are scored against your labels: accuracy, precision, recall, F1 and the confusion matrix.

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[Choose file](#) No file chosen

Loaded **10** patient notes from **evaluation_patients_unstructured.csv** (sample).

2 · Labelled structured data * (ground truth)

CSV with columns `patient_id`, `trial_id`, `actual_label` — `actual_label`: 1 = Potentially Eligible, 0 = Not Eligible, 2 = Insufficient Information.

[Choose file](#) No file chosen

Loaded **21** labels from **evaluation_labels_ground_truth.csv**.

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Which trials should accuracy be scored against? **All trials** checks every trial in the database. **Top-K** scores only each patient's K best-matching trials (by match score) — the way a coordinator actually reads the ranked list. Labelled pairs that fall outside a patient's top-K are skipped and reported.

[All trials](#)

[Top 3 trials](#)

[Top 5 trials](#)

[Top 10 trials](#)

The screenshot shows an actual top-5 scoped run: 13 labelled pairs scored, 8 skipped for falling outside a patient's top-5 matches (the count is printed next to the matrix), Accuracy / Precision / Recall / F1 at 100%, and the colour-coded confusion matrix. An expandable 'ML extraction details' panel lists the facts the NER found for each patient.

4.11 Per-pair predictions vs labels



Clinical Trial Qualifier

AI-Powered Clinical Trial Matching

HomeFind TrialsModel Evaluation

► ML extraction details (10 patients)

Predictions vs labels

PATIENT	TRIAL	PREDICTED	ACTUAL	MATCH
P001	CTRI/2024/01/062001	Potentially Eligible	Eligible	✓
P002	CTRI/2024/01/062001	Not Eligible	Not Eligible	✓
P001	CTRI/2025/08/064022	Not Eligible	Not Eligible	✓
P005	CTRI/2025/08/064022	Potentially Eligible	Eligible	✓
P005	CTRI/2024/01/062001	Potentially Eligible	Eligible	✓
P007	CTRI/2024/07/062744	Potentially Eligible	Eligible	✓
P001	CTRI/2024/07/062744	Not Eligible	Not Eligible	✓
P003	CTRI/2024/03/062318	Potentially Eligible	Eligible	✓
P004	CTRI/2024/08/062851	Potentially Eligible	Eligible	✓
P008	CTRI/2024/01/062001	Potentially Eligible	Eligible	✓
P009	CTRI/2025/04/063673	Potentially Eligible	Eligible	✓
P004	CTRI/2025/04/063673	Not Eligible	Not Eligible	✓
P010	CTRI/2024/11/063128	Potentially Eligible	Eligible	✓

Download results (CSV)

Clinical Trial Qualifier - Trial metadata: Clinical Trials Registry - India (CTRI)

Each scored row with a ✓: prediction equals the ground-truth label. The table downloads as CSV ('evaluation_predictions.csv') for offline inspection.

5. The tech stack - what each technology does and how

The app is a classic two-part web application: a React single-page app in the browser talking to a Python FastAPI server. The table summarises every piece; the paragraphs after explain the interesting ones with a worked example each.

Technology	Role in CTQ	How it is used here
DistilBERT extractive-QA (distilbert-base-uncased-distilled-squad)	Machine Reading Comprehension	Asks one targeted question per clinical field per note ('What is the patient's HbA1c level?')
Groq API (openai/gpt-oss-120b)	Translation + LLM review	Two jobs: (1) low-effort multilingual translation of non-Latin notes (Hindi + 12 scripts) to English
RapidOCR (ONNX) + PyMuPDF	OCR & PDF reading	Photos/scans of prescriptions → RapidOCR text; scanned PDF pages are rasterised with PyMuPDF
sentence-transformers (all-MiniLM-L6-v2)	Semantic embeddings	Encodes the patient profile text and each trial's criteria into 384-d vectors; cosine similarity
Custom rule engine (services/eligibility.py)	Idiosyncratic checking	Parses numeric cut-offs from criterion text ('HbA1c between 7.5 and 10.5', 'eGFR below 30')
FastAPI + Uvicorn	Backend REST API	/api endpoints: patient/analyze, patient/analyze-batch (evaluation), extract/document, cohorts
Pydantic	Data models + validation	PatientProfile, Medication, Trial and result schemas; coerces form/CSV values and guards against bad data
React 18 + Vite + react-router	Frontend UI	Home, Find Trials (two tabs + cohort), Results, Model Evaluation; axios API client; production-ready
pandas + openpyxl	Tabular parsing / Excel	Reads uploaded CSV/TSV/Excel with column aliasing; openpyxl also writes the cohort EDC
SQLite	Local storage	Stores evaluation runs; trials.json (32 CTRI/NCT-format trials, 200+ criteria) loads at startup
Playwright + matplotlib	Testing & figures	Automated end-to-end browser runs produced every screenshot in section 4; matplotlib renders

5.1 The ML NER step in detail - with the worked example

Input note (P001): “48-year-old male with Type 2 Diabetes Mellitus diagnosed 4 years ago. Height 170 cm, weight 82 kg. Increased thirst and fatigue. On stable Metformin 500 mg twice daily for the last 12 months. HbA1c 8.5 percent, eGFR 80, ALT 25 U/L. No history of liver disease. Does not smoke.” The engine asks the model a list of targeted questions and keeps a span only when it passes validation:

“What is the patient's HbA1c level?” → span “8.5” - accepted because the number sits next to the words 'HbA1c' (anchor check, word-boundary matched so 'alt' never matches inside 'heALThy').

“Does the patient have any history of liver disease?” → “No history of liver disease” → normalised to comorbidity 'Liver disease: No' - which the rule engine later uses to satisfy that trial's exclusion clause.

The diagnosis question initially returned “liver disease” from the negated sentence; a negation guard rejects spans found inside “No history of...” contexts, and a sentence-level fallback then lifts the real diagnosis 'Type 2 Diabetes Mellitus'.

Numbers already claimed by one field (height 170) cannot be stolen by another (BP), preventing cross-contaminated extractions; blood pressure must match a real '150/95' digit-slash-digit pair so 'ALT 26 U/L' cannot be read as a BP of 26.

Output facts: age 48, gender Male, condition Type 2 Diabetes Mellitus, duration 48 months, Metformin 500 mg twice daily (12 mo), HbA1c 8.5, eGFR 80, ALT 25, height/weight → BMI 28.4, Liver disease: No, smoking Never.

5.2 The rule engine in detail - where safety lives

For the same patient against trial CTRI/2024/01/062001 (T2DM add-on to metformin), the engine parsed the criterion text and produced: “Patient age 48 is within the required range 30-65 years”; “HbA1c 8.5 satisfies inclusion criterion ‘HbA1c between 7.5 and 10.5 percent’”; “Current medication Metformin matches trial requirement of stable metformin ≥ 3 months (12 months recorded)” → verdict **Potentially Eligible**. Against CTRI/2025/08/064022 (an obesity trial) it produced “BMI 28.4 does not satisfy inclusion criterion ‘Body mass index between 30 and 45’” → **Not Eligible**. Verdict policy: any hard exclusion wins

(Not Eligible); missing inclusion-side facts (Insufficient Information); missing exclusion-side facts only (stays Potentially Eligible, gaps listed). A criterion like 'ALT above 3 times the upper limit of normal' is recognised as a multiplier phrase, not a literal cut-off of 3.

5.3 The multilingual translation layer (Hindi today, 13 scripts)

[illegible]

Detect. `services/translate.py` scans the note's alphabetic characters against Unicode script ranges (Devanagari, Bengali, Gurmukhi, Gujarati, Odia, Tamil, Telugu, Kannada, Malayalam, CJK, Cyrillic, Arabic, Hebrew, Greek). If $\geq 5\%$ belong to a non-Latin script, the note is flagged; detection is instant and offline.

Translate. The note goes to the Groq LLM with a strict medical-translation prompt (keep every number, unit and lab name exactly as written; ■■■■■■■■■■ → Metformin; preserve line breaks and [ID] markers) with low reasoning effort to save tokens.

Extract. The English text flows through the untouched, validated NER pipeline - no change to the extraction behaviour that scored 100% on the English sample.

Fail safe. Translation shares the LLM rate-limit breaker. With no API key, a 429, or any failure, the original text is used and the UI explains what happened - extraction never breaks because translation did.
Hindi-verified result: age 48, male, diabetes, Metformin 500 mg twice daily, HbA1c 8.5, creatinine 1.1, eGFR 80, BP 140/85, Never/Never lifestyle - the complete profile, from pure Hindi input.

5.4 The evaluation pipeline in detail

POST /api/patient/analyze-batch receives the unstructured patients and the labels; each note is re-extracted by the ML NER (explicit fields win), every trial is checked for every patient via the single-patient code path, and the labelled pairs are scored: 1 = eligible, 0 = not eligible, 2 = insufficient (excluded from the binary matrix, tracked separately). Per-trial and per-patient accuracy breakdowns, the extraction log and a downloadable predictions CSV come back with the metrics. The frontend's 'Load bundled sample data' button simply calls /api/evaluation/sample-data, which serves exactly these CSVs from sample_data/.

5.5 One request, end to end

Form submit (or file upload → ML NER) - buildPayload() structures the JSON - POST /api/patient/analyze
 - build_profile() normalises it into PatientProfile - trial_retrieval pre-filters/annotates all 32 trials -
 embed_texts() embeds the profile and each trial's criteria - cosine similarity ranks them (display order; all
 are checked) - evaluate_eligibility() runs the rule engine per trial - verdicts merge - AnalyzeResponse
 returns - the Results page renders verdict cards with reasons and the CSV download. Typically 5-20
 seconds for a full 32-trial sweep per patient on CPU.

6. What can be improved next

Domain-specific NER. Swap or ensemble DistilBERT-QA with BioBERT / PubMedBERT fine-tuned on CTRI criteria for higher recall on rare entities and lab names.

Dose-threshold rules. The rule engine can parse 'HbA1c <= 9.0' but not yet 'metformin at least 1500 mg/day'; the dose field on medications makes that check possible.

Real registry data. Point the retrieval layer at a live CTRI export or ClinicalTrials.gov (NCT) JSON - the trial import path already accepts the same schema.

Doctor feedback loop. Human-in-the-loop corrections that feed back into the rules and the evaluation set.

More languages + speech. Hindi notes ship today (13 scripts detected, translated before extraction); regional speech-to-text dictation and ICD / LOINC coding come next.

Accounts + investigator dashboard. Save patient histories, track which trials were contacted, record enrolment outcomes - turning CTQ from a screener into a workflow tool.

PostgreSQL + deployment. Swap SQLite for PostgreSQL and deploy the API and frontend for multi-user access.

7. Deployment - live on Render

CTQ runs publicly at <https://ctq-clinical-trial-qualifier.onrender.com> (Docker deploy, auto-redeploys on every push to master). To fit the free tier's 512 MB RAM, the DistilBERT NER is exported to ONNX int8 at Docker build time (66 MB self-contained artifact; onnxruntime with single-thread execution and no arena growth - torch is never imported on that path), and embeddings run in a deterministic hashed mode. The Groq key is injected as a sync:false environment variable. Known free-tier limits: after 15 minutes idle the server sleeps (~50 s cold start on the next request), CPU throttling means very large multi-note uploads may time out, and the Groq daily token cap (~200k TPD) occasionally pauses LLM translation/review until the breaker resets - the rule engine keeps verdicts flowing either way.

CTQ - Clinical Trial Qualifier. Backend: FastAPI + SQLite + DistilBERT extractive-QA ML NER + Groq translation/review + sentence-transformers + RapidOCR. Frontend: React + Vite. Deployed on Render. Trial metadata modelled on the Clinical Trials Registry - India (CTRI). All demo data is synthetic; no real patient data is used.